

Source: [SageJournals](#)

The method of diet monitoring that time

- Unsustainable: Through pen and paper recordings and manual nutrient coding respectively.

Based on FGD:

- Keywords used on App Store: diet, nutrition, weight loss, calorie tracker, and food recognition.
- Premium features in certain diet applications that require subscriptions were not examined.

Table 3. Findings from the focus on needs in a mobile diet application.

	Dietitians	Potential users	Dietitians and potential users
Personalised recommendations	-Provide health tips	-Reduce notification frequency -Goal setting	-Provide individualised dietary recommendations *Provide disease-based information with appropriate food choice recommendations *Collaboration with other health-related applications to provide education or rewards when goals are achieved -Provide access to support groups with specific interests
Food logging methods	*Log foods through speech recognition (audio)	-Provide a list of commonly logged foods for faster data entry *Location-based food choice recommendations -Option to edit past food log	-Food recognition and *automated portion size estimation -Reduce steps required for data entry to reduce user burden *Barcode scanner to display nutrient information of packaged foods -Provide reminders to log meals
Food database and portion size estimations	-Use a localised and credible food database *Allow researchers to have access to update database		-Comprehensive food database that *includes nutritional supplements -Ease portion size estimations -Options to select portion consumed -Provide portion estimation guide and portion selection options in grams and household measures
Nutrient summary	-Provide interpretation of reference and recommended values	-Limit micronutrient information as it may seem overwhelming *Display classification of foods according to nutrient groups -Present information using traffic light system instead of numbers *Option to track water intake	-Display breakdown of key nutrients consumption and nutrient analysis through graphs, pie charts, bar charts -Display the history of intake *Option to view diet history from a website to view long-term trends
General and technical issues	-Contains misleading information *Separate dashboards for researchers (more detailed) and users (simple) *Multiple language availability *Allow two-way communication between healthcare professionals and user	-Tiring to key in every meal -Ensure a pleasant user experience (minimise delays and crashes) -Option for offline availability *Provide cooking-related information (local recipes, healthier alternatives) to educate users	-Utilising large, clear, and neat icons -Less word usage -Provide tutorial

*Suggestions that were not incorporated into MuFitCam: FGD: Focus Group Discussion

1. Personalized recommendations:
 - The most common answer was providing individualised dietary recommendations such as recommended daily calorie and nutrient intakes and the link between users' biochemical data with their dietary intake.
 - Dietitians: Providing users with health tips relevant to their current diet intakes.
 - Potential users: Reduce notification frequencies through the freedom to set the timing and frequencies of the alert as compared with currently available diet applications.

2. Food logging methods

- Quite fond of using newer methods of food logging
- Prefer if calories and nutrients of the foods could be automatically measured from a single photo.

*No one has yet to implement this feature in the real world successfully.

- Other suggestions: Speech recognition, a list of commonly consumed foods, location-based monitoring, and barcode scanners.

3. Food database and portion size estimations

- Preferred the diet app to contain a comprehensive food database that includes localised data and nutritional supplements.
- Dietitians: Allow researchers to update the food and nutrient databases whenever required.
- Most agreed that portion size guides should be provided to ease the portion size estimations and offer end users the freedom to select their portions in terms of grams or household measures instead of fixed standardised portions.

4. Nutrient summary

- Predominantly requested for visual breakdown of key nutrients, including calories, macronutrients, and micronutrients.
- Other suggestions: To limit micronutrient information as it may seem overwhelming to users who are just getting started.
- Potential users: Presenting information using the traffic light system instead of numbers.

So instead of overwhelming users with:

“Sodium: 1450mg”

the app could say:

● High sodium intake for this meal

- Dietitians: Provide interpretations of reference and recommended values to guide users in understanding the nutrition data better.
- Dietitians: The ability to refer or view what the users had consumed in the past would also be a pleasant feature to be added.

5. General and technical issues

- Utilise large, clear, and neat icons with minimal word usage in addition to provide users with a tutorial guide.
- Dietitians: current diet applications containing misleading information

- Potential users: talked about the inconvenience of having to key in every single meal.

Competitors:

Through LR:

- *Healthy365 (Food AI)*
- *Im2Calories*
- *DietCam*: instant generation of estimated calories and nutrients of a dish from a single video. (If this feature is done accurately and covers an extensive list of food items in the real world, it would remove a huge barrier in dietary monitoring from manually estimating food portions of each food item in a dish.)
- *Snap-N-Eat*

App Stores (mainly developed by private app development companies) :

*Dietitian: commercially available apps in app stores, it is mainly developed by private app development companies. It may not be necessarily developed by experts in the field and that may result to provision of inaccurate information in the app

- *MyNutriDiari 2*
- *MyFitnessPal*
- *FatSecret*
- *Lose It!*
- *Calorie Mama*
- *DietCamera AI* (Korean)
- BlueLoop
- Bezy T2D
-

MyFitnessPal, FatSecret, and LoseIt!:

- Provide users with a barcode scanner which is very effective for retail items.
- food recognition features are also still in the early stages, requiring human input to confirm the detected food items and manual portion size entries.
- Regarding the domains featured, none of the other inspected applications provided a diet quality score or any other measures to help users understand or summarise their daily nutrient data in a way that is easier to understand.
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Methods of developing the MyDietCam

- Targeted at healthy Malaysian adults.
- Focus group discussions (FGD) were carried out among dietitians and potential users to determine ideal features in a diet application.
- Conducted 6 FGD.

- The first three were conducted among dietitians while the other three were conducted among potential diet app users.
- 30 participants were recruited from a local university to log their meals through MyDietCam for six days and submit the Malay mHealth Application Usability Questionnaire (M-MAUQ) at the end of the study.
- Adopted a multidisciplinary approach involving a collaborative team of researchers from the field of dietetics, behaviour change, and artificial intelligence.
-

Results from FGD

Led to the implementation of the main features:

- Individualised recommendations
- Food logging through food recognition to reduce steps for data entry
- Provide detailed nutrient analyses through visuals.

*Scored 5.13/7 reported from M-MAUQ = Acceptable

MyDietCam Features

MyDietCam Features	Description
Tutorial	In-app link to step-by-step YouTube tutorial videos
Goal setting	Enable users to set their calorie and weight goals
Food logging	(a) Food recognition- The user can log multiple food items through a single image (b) Manual text entry- The user can type every individual food item manually in the search bar Freedom to select own portion size with various selection options (portion size estimation guide available)
Food intake summary	Show the list of foods that have been logged according to each meal Calorie intake displayed as a progress bar Display the calorie content of each meal Edit previous logs (portion size changes, deleting food items, moving food items into a different meal)
Nutrient summary	Contain a calorie intake progress bar Distribution of calories and percentage logged across meals Macronutrient (carbohydrate, protein, and fat) distribution Micronutrient (iron, calcium, sodium, vitamin C) and fibre intake progress Diet quality score Display reference intake or ranges for each nutrient
Diet quality score	Generated based on the Malaysian Healthy Eating Index (M-HEI)
Diet history	Contains a bar chart showing the history of calories logged per day Enable users to view and edit their past intakes for up to three months
Exporting report	Users can export their diet data in a PDF report to be shared with researchers or healthcare professionals Report contains all food and nutrient data and analyses of the selected days

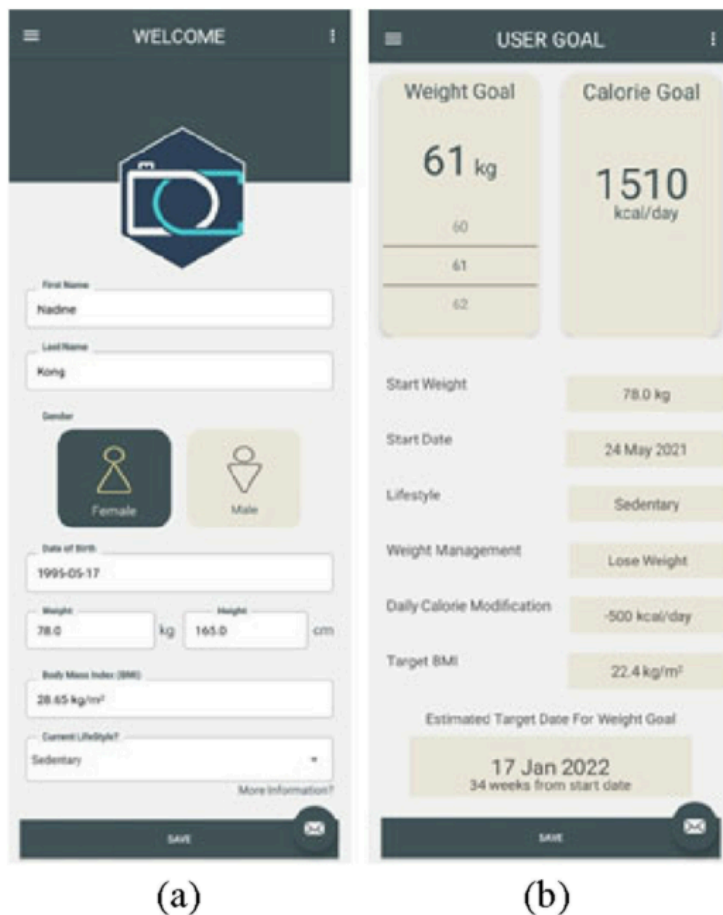


Figure 2. Mydietcam screenshots of user profile and user goal.

(a) User Profile

- Enter their gender, age, weight, height, and activity level.
- Would partially determine the calorie group they will be placed in.

(b) User Goals

- users who intend to lose or gain weight through calorie modifications.
- will be able to select their weight goal, the intensity of their calorie modifications, and the estimated target date to achieve their weight goal.

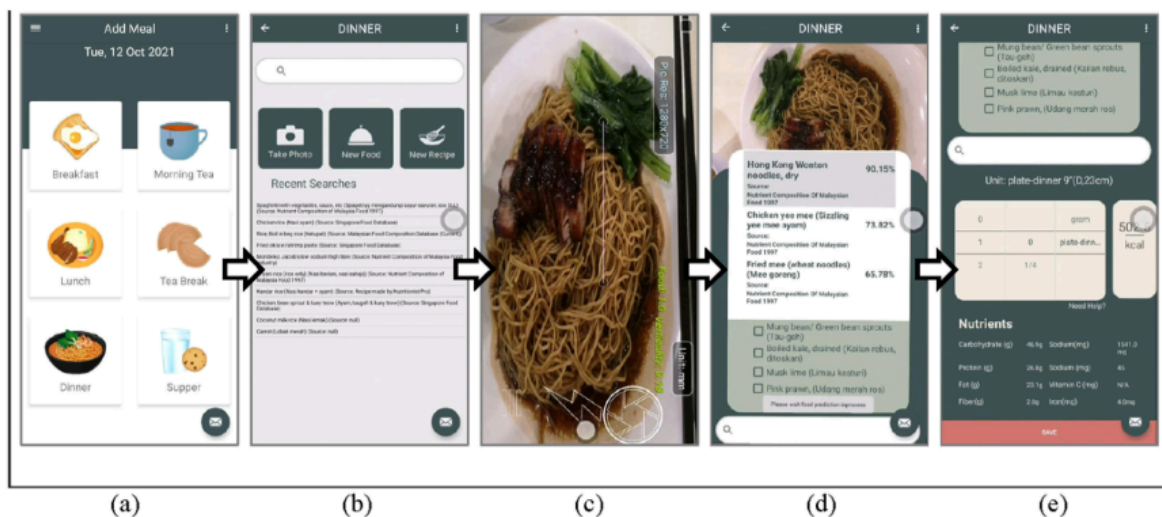


Figure 3. Flow of MyDietCam food logging process through image recognition.

Food Logging Process

- user selects one of the six meals from the dashboard
- tap on the camera icon
- tap on camera shutter button after ensuring food item is in the frame
- image will be sent to an online recognition server where the user will be presented with the top three predictions
- selects the correct prediction
- adjust portion size and save food item at the bottom of the screen.



Figure 4. Mydietcam screenshots of (a) food diary (b) nutrient summary (c) diet quality and (d) history.

- Food Diary:** shows the food intake summary (Food Diary) screen where users can see their calorie intake progress. The bar will turn red if calorie intake exceeds the calorie goal. The list of food items and their calorie values for each meal is presented on this screen. From here, users can click on any food item to modify the log by deleting the food, changing the portions, or moving the item to a different meal.
- Nutrient Summary:** colour-coded pie charts illustrate the distribution of calories across the different meals of the day and the macronutrient (carbohydrate, protein, and fat) distributions. Recommended values are also shown on the screen for the user's reference. Information on micronutrients (sodium, calcium, iron, and vitamin C) and fibre are presented in a table that shows their current intake, reference goal, and the difference between the values.
- diet quality screen:** breakdown of the intake progress of each food group and nutrient required for M-HEI calculation.
- History:** shows the diet history that displays the user's past 7-day calorie intake by default. Users can enter a customised date range to change the bar chart display. The bottom of the screen shows the calculated values of the highest, lowest, and average calorie intake for the selected date range.

M-MAUQ

MAUQ Statements (English version) ¹¹		Mean score (SD)
Overall		5.13 (1.15)
Domain 1: Ease of Use		5.14 (1.19)
1	The app was easy to use.	4.63 (1.52)
2	It was easy for me to learn to use the app.	5.07 (1.41)
3	The navigation was consistent when moving between screens.	5.47 (1.33)
4	The interface of the app allowed me to use all the functions (such as entering information, responding to reminders, viewing information) offered by the app.	5.47 (1.30)
5	Whenever I made a mistake using the app, I could recover easily and quickly.	4.80 (1.45)
Domain 2: User Interface and Satisfaction		5.10 (1.22)
6	I like the interface of the app.	4.70 (1.51)
7	The information in the app was well organised, so I could easily find the information I needed.	5.03 (1.00)
8	The app adequately acknowledged and provided information to let me know the progress of my action.	5.47 (1.46)
9	I feel comfortable using this app in social settings.	5.47 (1.41)
10	The amount of time involved in using this app has been fitting for me.	5.27 (1.66)
11	I would use this app again.	4.77 (1.57)
12	Overall, I am satisfied with this app.	4.80 (1.58)
Domain 3: Usefulness		5.17 (1.17)
13	The app would be useful for my health and well-being.	5.93 (1.36)
14	The app improved my access to healthcare services.	5.60 (1.54)
15	The app helped me manage my health effectively.	5.43 (1.45)
16	This app has all the functions and capabilities I expected it to have.	5.03 (1.54)
17	I could use the app even when the Internet connection was poor or not available.	4.30 (1.60)
18	This mHealth app provides an acceptable way to receive healthcare services, such as accessing educational materials, tracking my own activities, and performing self-assessment.	5.27 (1.57)
M-MAUQ: Malay mHealth Usability Questionnaire; ^a scores range from 1(low usability) to 7 (high usability), scores above 4 are considered acceptable usability; ⁴⁰ mHealth: mobile health.		

Key Issues:

Table 8. Key issues identified from usability testing.

Key Issues	Frequency (%) ^a
Limitations in food database options and search functionality (require specific keywords and spelling) which led to users not being able to find the food item	9 (30)
Application crashes and lags too frequently (delayed transitions and click responses)	12 (40)
Inaccurate food recognition	5 (17)
Difficulty in estimating portion ingredients for outside foods	5 (17)
Unclear navigations (portion size selection, modifying past intakes)	7 (23)
Unable to use without internet	3 (10)
Users constantly forget to log food items	3 (10)

^a Frequency % is calculated based on the number of participants ($n = 30$).

Suggestions provided through usability testing:

Suggestions for Improvement	Frequency (%) ^a
Improve food choices in database - Adopt unofficial list to allow more choices when logging foods - Allow users to add own food/ recipes	10 (30)
Fix minor errors Redirect users back to Food Diary screen after editing	3 (10)
Improve food recognition accuracy	5 (16)
Ease navigations	4 (13)
Ease portion size estimation process - Add reference photos and volume for household measures (cups/ bowls/plates) - Provide more portion size selections	6 (20)
Provide educational materials - Include explanations of functions of each macro- and micronutrients - Include explanations of diet-disease relationships - Provide reference on the recommended amounts for each food group in diet quality screen	4 (13)
Alternative food logging methods - Enable uploading pictures from phone gallery - Allow barcode scanning of commercial food item - Allow users to add own food items	3 (10)
Additional features - Add motivational note after every log - Notifications and reminders	1 (3) 4 (13)
Logging of foods according to meals may not be feasible for users who have inconsistent meal times	1 (3)
^a Frequency % is calculated based on the number of participants (<i>n</i> = 30).	

Other insights:

- On top of that, potential users suggested labelling foods with certain health risks such as foods with high purine content, high phosphate, and high glycaemic index, and providing cooking-related information such as healthy recipe modifications.
- Most of the suggestions were focused on the diseased population, which is not MyDietCam's target population. It is crucial to stay on track with the agreed application outcomes designed for the target population.
- there were also recommendations to provide two-way communication between healthcare professionals through the application.
- The M-MAUQ was used in this study instead of other scales such as the mobile app rating scale (MARS)⁵⁶ or the user mobile app rating scale (uMARS),⁵⁷ as the MARS was developed to target experts in the field to identify high quality apps in their field.

Strengths:

- first Malaysian mobile diet application incorporated with food recognition technology aimed at general healthy adults.
- instant generation of an overall diet quality score

Limitations:

- small sample conducted in a university setting.
- Although the age range of the participants covered MyDietCam's targeted age groups, the participants all comprised highly educated individuals with a nutrition background.
- participants were all residing in the Klang Valley, and thus do not generalise to residents from other regions in Malaysia where different cultures and cuisines are observed.
- premium or paid features from commercial-based applications were not examined during the inspection of previously available applications
- it requires a stable internet connection to log food items as it is connected to an online server.
- The food and portion options from the food database may be outdated but it is currently being updated.
- keeping food databases updated is an ongoing challenge, especially with the rate of new commercial food products.

Suggestions based on usability testing:

- to expand the food database to include more food items.
- A wide variety of food choices enables users to log their diets more accurately instead of logging items closest to what they have consumed.
- food database to be updated and tested in a diversified population with different age groups and education levels.

- The food recognition accuracy has to be further improved through more training and testing especially in the local context.

Other Competitors / Apps

Feature	mySugr	Glucose Buddy	CGM Apps (Dexcom/Libre)	Fooducate / CalorieKing	BlueLoop	Bezy T2D
Primary Focus	Daily logging & Gamification	Data integration & Medical reports	Real-time glucose streaming	Carb & Calorie counting	Pediatric care & Family sharing	Community & Emotional support
Best For	Type 1 & 2 (All ages)	Users who share data with doctors	Intensive insulin users	Weight loss & Meal planning	Parents of children with T1D	Type 2 Peer support
Unique Selling Point	The "Diabetes Monster" & Bolus Calculator	Automatic syncing with many meters	No finger-pricks (Real-time trends)	Letter grades for food health (A-F)	Real-time alerts for caregivers	Medically moderated forums
Cost Structure	Free (Pro version available)	Free (Premium for full reports)	App is free; Sensor hardware is costly	Free (Subscription for allergens)	Free	Free
Connectivity	Accu-Chek & ReliOn meters	Dexcom & Apple Health	Integrated with Pumps/Sensors	Apple Health / Google Fit	Online portal for school nurses	Social platform

Understanding the Key Players

- **mySugr:** The "Personal Assistant." It focuses on making the daily chore of logging less boring through gamification. Its "Diabetes Monster" reacts to your logs, helping you "tame" your condition.
- **Glucose Buddy:** The "Data Hub." It is designed for users who want a comprehensive medical log. It's less about games and more about generating clean reports you can hand to your doctor.
- **FreeStyle Libre & Dexcom:** The "Hardware Powerhouses." These aren't just apps; they are ecosystems. They use physical sensors on your body to provide 24/7 data, replacing the need for most finger-prick tests.
- **Fooducate & CalorieKing:** The "Nutrition Coaches." These focus specifically on the *input* side of the equation. They help you understand exactly how many carbs or hidden sugars are in a specific barcode.
- **BlueLoop:** The "Coordination Tool." Designed specifically for families. It ensures that when a child logs a sugar level at school, the parent and the school nurse both get an instant notification.
- **Bezy T2D:** The "Support Group." It's not a tracker; it's a social network. It's for when you need to talk to someone who actually understands the fatigue of living with Type 2.

Competitive Analysis

Platform	Core Focus	Strengths	Weaknesses / Gaps	Opportunity for GlucoLens
MyNutriDiari 2	Malaysian calorie tracking	Government-backed, Malaysian food database	Manual logging, less AI automation, limited personalization	GlucoLens automates meal recognition through photos and provides diabetic-specific feedback
MyFitnessPal	Global fitness & calorie tracking	Huge food database, strong brand recognition	Western-food centric, manual entry fatigue, generic health recommendations	GlucoLens specializes in Malaysian foods and clinical diabetic monitoring
FatSecret	Diet & calorie logging	Community support, barcode scanning	Weak localization for Malaysian cuisine	GlucoLens focuses on local dishes like nasi lemak, roti canai, mixed rice
Lose It!	Weight loss & calorie goals	User-friendly UX, gamification	Primarily weight-loss oriented, not clinical nutrition	GlucoLens targets medical nutrition therapy for T2DM
Calorie Mama AI	AI food recognition	Strong AI image recognition	Limited Southeast Asian food accuracy	MyDietCam IP already emphasizes Malaysian food recognition as a core differentiator
DietCamera AI	Korean AI diet analysis	Advanced food-photo analysis	Korea-centric food dataset and nutrition habits	GlucoLens can dominate the underserved Malaysian diabetic nutrition niche
BlueLoop	Diabetes management for families	Care coordination, reminders	Not nutrition-AI focused	GlucoLens integrates meal intelligence + dietitian workflow

Bezzzy T2D	T2D support community	Peer support ecosystem	Limited food recognition and nutritional analytics	GlucoLens positions itself as a clinical nutrition companion instead of social platform
mySugr	Diabetes tracking	Strong glucose logging ecosystem	Requires manual food tracking	GlucoLens reduces logging friction using image-based food recognition
Glucose Buddy	Blood sugar management	Comprehensive diabetic tracking	Nutrition guidance not localized	GlucoLens combines Malaysian food AI + nutrition recommendations
FreeStyle Libre & Dexcom	Continuous glucose monitoring (CGM)	Real-time glucose data, strong clinical credibility	Hardware-dependent, expensive, limited meal interpretation	GlucoLens complements CGM by explaining <i>why</i> glucose spikes happen through meal analysis
Fooducate & CalorieKing	Nutrition education	Strong nutrition database and food scoring	Not optimized for Malaysian diets or diabetes workflow	GlucoLens adds cultural relevance + AI-powered personalized recommendations

Strategic Positioning Map

You can summarize the market gap into **4 major categories**:

1. Global Fitness Apps

Examples:

- MyFitnessPal
- Lose It!
- FatSecret

Gap:

Good at calorie counting, weak at:

- Malaysian foods

- diabetic personalization
 - healthcare integration
-

2. AI Food Recognition Apps

Examples:

- Calorie Mama
- DietCamera AI

Gap:

Strong AI, but:

- mostly Western/Korean datasets
 - not clinically integrated
 - limited diabetic monitoring
-

3. Diabetes Monitoring Apps

Examples:

- mySugr
- Glucose Buddy
- BlueLoop

Gap:

Strong glucose logging, weak at:

- meal intelligence
 - automated nutrition analysis
 - culturally relevant food guidance
-

4. CGM Ecosystems

Examples:

- Dexcom
- FreeStyle Libre

Gap:

Excellent glucose sensing, but:

- do not interpret Malaysian meals
- expensive hardware dependence

- weak dietary behavioral coaching

Suggested “Competitive Advantage” Slide

You can structure the slide like this:

Capability	Global Fitness Apps	Diabetes Apps	AI Food Apps	Glucolens
Malaysian Food Recognition	✗	✗	⚠ Limited	✓
AI Meal Photo Analysis	⚠ Partial	✗	✓	✓
Diabetes-Specific Recommendations	✗	✓	✗	✓
Dietitian Dashboard	✗	⚠ Limited	✗	✓
Behavioral Nudges	⚠ Generic	⚠ Generic	✗	✓
Local Clinical Relevance	✗	✗	✗	✓

Grants

Core Funding Narrative

Glucolens addresses Malaysia's growing Type 2 Diabetes burden by transforming everyday meal photos into clinically actionable nutrition insights using Malaysian food-recognition AI. By integrating the MyDietCam IP with personalized diabetic guidance, Glucolens reduces manual dietary logging barriers and enables continuous nutrition monitoring between clinical visits.

This directly aligns with:

- AI & Digital Health
- Preventive Healthcare
- Community Health Empowerment
- University IP Commercialization
- Student Innovation & Entrepreneurship

Key Problem Justification

The Healthcare Gap

Malaysia has approximately:

- **3.9 million diabetics**, according to your PRD.

A major challenge is:

- patients only meet dietitians periodically,
- but make daily dietary decisions independently.

Your PRD already frames this strongly:

“Between consultations, patients consume an estimated 360 meals unsupervised.”

That is a VERY strong grant narrative because it shows:

- healthcare resource limitation,
- unmet monitoring gap,
- scalable digital intervention potential.

Why Existing Solutions Are Insufficient

Current apps:

- focus on calorie tracking,
- are Western-food centric,
- require manual logging,
- lack Malaysian clinical relevance.

Meanwhile MyDietCam already has:

- Malaysian food recognition,
- behavioral science integration,
- nutrition analysis,
- healthcare support capabilities.

This creates a strong “local innovation solving local healthcare challenges” narrative.

Tailored Justification by Grantor

1. Pusat Keusahawanan Pelajar Universiti Malaya

What They Care About

- student entrepreneurship,
- commercialization of innovation,
- employability,
- UM innovation ecosystem.

Strong Justification

Funding GlucoLens supports the commercialization of Malaysian university-linked healthcare innovation while empowering students to transform research IP into real-world impact. The project demonstrates interdisciplinary collaboration between AI, healthcare, and entrepreneurship, aligned with Universiti Malaya’s goal of nurturing innovation-driven student founders.

Key Value

- UM-linked innovation
- student startup potential

- commercialization readiness
- high-impact social innovation

2. Cradle Fund

What They Care About

- scalable startups,
- commercialization,
- innovation readiness,
- high-growth technology ventures.

Strong Justification

Glucolens demonstrates strong commercialization potential through its AI-powered Malaysian food-recognition technology and scalable SaaS healthcare model. With the underlying IP already at TRL 6–7, grant funding would accelerate pilot deployment, clinical validation, and customer acquisition, bridging the gap between research innovation and market adoption.

Key Value

- TRL 6–7 readiness
- scalable SaaS pathway
- defensible Malaysian dataset
- healthcare AI market opportunity
- recurring revenue potential

Suggested Milestones

- 3 pilot clinics
- 100 diabetic users
- validation accuracy benchmarking
- dietitian dashboard MVP
- clinical engagement metrics

3. UM Startup Community

What They Care About

- startup ecosystem growth,
- founder capability,
- innovation culture,
- scalable student ventures.

Strong Justification

Glucolens represents a high-potential student-led deep tech startup built on real healthcare challenges in Malaysia. The project combines AI, digital health, and behavioral science to address a growing chronic disease burden while creating opportunities for startup incubation, industry collaboration, and long-term venture scalability.

Key Value

- student founder ecosystem
- AI startup exposure
- deep tech commercialization
- social entrepreneurship

4. Sunway University

What They Care About

- healthcare innovation,
- sustainability,
- SDG alignment,
- translational research.

Strong Justification

GlucoLens contributes toward Sustainable Development Goal 3 (Good Health and Well-being) by enabling accessible and preventive diabetes nutrition management using AI technology. The platform promotes healthier dietary behavior and supports early intervention strategies, particularly relevant in Southeast Asia's rising non-communicable disease landscape.

SDG Alignment

- SDG 3: Good Health & Well-being
- SDG 9: Industry, Innovation & Infrastructure

Key Value

- preventive healthcare
- AI for public health
- research commercialization
- regional healthcare scalability

5. Junior Chamber International Malaysia

What They Care About

- community impact,
- youth leadership,
- social innovation,
- scalable societal benefit.

Strong Justification

Glucolens empowers communities through accessible digital healthcare innovation that encourages healthier lifestyles and proactive diabetes management. By reducing barriers to nutrition monitoring and promoting behavioral change, the solution has the potential to improve quality of life among underserved and at-risk populations.

Key Value

- community empowerment
- youth-led innovation
- scalable social impact
- preventive health awareness

Slide 1: The Problem — The "Quarterly Gap"

Mentor's Note: Use dramatic numbers and emphasize the "pain."

- **Visual:** A large "360" in the center. On one side, a tiny icon of a doctor. On the other, 360 icons of plates.
- **Content:**
 - **The Statistic:** Malaysia has 3.9 million diabetics. They see a dietitian 4 times a year.
 - **The Pain:** In between those visits, they eat **360 meals alone** with zero guidance.
 - **The Consequence:** 87% of patients have poor blood sugar level (HbA1c) control, contributing to a **RM 4.1B annual healthcare cost** in Malaysia.
***need to fact check**
- **The Hook:** "We aren't just an app; we are the safety net for those 360 unsupervised decisions."
- T2DM is not curable. They have to be mindful of their eating habit for the rest of their life, even if it after unsubscribe from our apps.

Slide 2: The IP — MyDietCam (The Foundation)

Mentor's Note: Focus on what the IP does and how it's the "secret sauce."

- **Visual:** .
- **Content:**
 - **Local Accuracy:** Powered by IP from Prof Moy (UTP), specifically trained on **Malaysian and SE Asian datasets**.
 - **The Edge:** Global apps (like MyFitnessPal) fail at local dishes like Roti Canai or Char Kway Teow. MyDietCam identifies them with high precision.
 - **Core Feature:** Meal photo analysis + automated portion estimation.
 - **** make sure core feature mentioned ada dalam glucoLens**

Slide 3: The Solution — GlucoLens & The 9 Agents

Mentor's Note: Introduce the "Malaysian Personalized Dietitian in your Pocket."

- **Visual:** A diagram showing the **9-Agent LangGraph Architecture**.
- **Content:**
 - **** need to mention verbally that we revamp the existing workflow into agentic flow, why choose**
 - **Beyond Tracking:** We don't just log; we **intervene**.
 - **The Agents:** Mention **Agent 3 (Clinical Personalization)** for drug-food interactions and **Agent 8 (Misinformation Debunker)** to fight TikTok health myths.
 - **Closing the Loop:** **Agent 6** generates an automated weekly PDF brief for the dietitian, turning a 20-minute recall-based chat into a data-driven clinical consultation.

Slide 4: The Market & Business Model (B2B2C)

Mentor's Note: Explain the subscription and why hospitals/insurers pay.

- **Visual:** A flowchart showing GlucoLens $\rightarrow$ Hospital $\rightarrow$ Patient.
- **Content:**
 - **B2B Licensing:** Selling to **2,000+ private clinics** and hospitals.
 - **B2C Subscription:** Freemium model for patients at RM 15–25/month.
 - **** add a table - subscription plan dalam business plan tu + breakeven analysis + the goal**
 - **Strategic Alignment:** We save insurers and the government money by reducing diabetic complications (the RM 4.1B problem).
 - **The Goal:** Target 25,000 active patients by Year 3, reaching a projected **RM 7.1M revenue**.

Slide 5: The Team — Technical & Financial Prowess

Mentor's Note: Highlight your CS background and financial minoring.

- **Visual:** Team photos with logos of UTP/UM and your tech stack (React, FastAPI, LangGraph).
- **Content:**
 - **Deep Tech Capability:** We built the full working prototype in just **72 hours**.
 - **Multidisciplinary:** Computer Science & Engineering students with **minors in Finance/Management**.
 - **** include work with agentic ai before**
 - **Experience:** Proven ability in fundraising and community project management (MPU4, Technopreneurship).

Slide 6: The Ask — RM 500K for Clinical Validation

Mentor's Note: "Shark Tank" style pitch. Be specific.

- **Visual:** A pie chart showing the **RM 500K Seed Funding** breakdown.
- **Content:**
 - **40% Product & Engineering:** Production-ready integration of MyDietCam.
 - **20% Clinical Validation:** Pilot with 100+ patients at Sunway Medical or KPJ to track HbA1c outcomes.
 - **10% Regulatory:** Navigating MDA (Medical Device Authority) classification.
- **Strategic Partners:** We are seeking introductions to **MOH Malaysia/KOSPEN** for clinical data integration

OTHER INSIGHTS

Prompt:

Could you help me to find % dietitians struggle to keep track of their patients blood level, diets, others that related to come up with their insights?

Also could you provide me the number that show the rising trends among Malaysian s that wanting to adopt healthy diet in social media?

Please provide the reliable source okiee

Answer:

Yes — and actually this is VERY GOOD material for your grant/commercialization narrative because now you can support your story with:

1. rising health awareness,
2. social-media-driven wellness behavior,
3. and gaps in clinical nutrition monitoring.

But there's one important thing:

There is NO reliable national statistic saying:

“X% of dietitians struggle to track patient diets.”

That exact metric is difficult to find because:

- dietitian workflow studies are usually qualitative,
- and many healthcare workflow pain points are discussed in journals/interviews rather than national percentages.

So instead of forcing an unreliable number, you should use:

- credible workflow evidence,
- healthcare workload evidence,
- and your own problem framing.

That is MUCH safer and more professional.

1. Strong Clinical Monitoring Problem Evidence

Your own PRD already provides a VERY strong insight:

“A typical dietitian sees each patient once per quarter. Between consultations, patients consume an estimated 360 meals unsupervised.”

This is actually excellent because it implies:

- lack of continuous monitoring,
- fragmented dietary tracking,
- and difficulty generating longitudinal patient insights.

You can frame it as:

“Dietitians often rely on infrequent consultations and self-reported food recall, limiting their ability to continuously monitor patient dietary behavior and correlate it with glucose outcomes.”

This is clinically accurate and defensible.

2. Reliable Evidence About Health Awareness Trends in Malaysia

National Healthy Lifestyle Statistics

A nationally representative Malaysian study based on the National Health and Morbidity Survey (NHMS 2019) found:

Only 30.6% of Malaysians practiced at least four healthy lifestyle behaviors.

This is VERY useful because it shows:

- growing awareness is still insufficient,
- lifestyle intervention opportunity remains huge.

You can use:

“Despite rising health awareness, only 30.6% of Malaysian adults met at least four healthy lifestyle behaviors according to NHMS 2019.”

3. Strong Evidence That Social Media Is Driving Health Behavior

TikTok & Social Media Penetration

According to Statista:

TikTok was downloaded more than 10 million times in Malaysia during 2024, with the majority of users aged 18–34.

This is VERY important because:

- your target users are digital-first,
 - wellness culture is social-media driven,
 - your adoption channel becomes believable.
-

4. Reliable Evidence That Nutrition Content Is Exploding Online

A 2025 nutrition-content study reported:

Health & wellness influencers accounted for 32% of nutrition-related TikTok creators.

You can use:

“Nutrition and wellness content on TikTok is increasingly driven by health influencers, reflecting growing consumer interest in healthier lifestyles and dietary awareness.”

5. VERY Strong Government Validation

This is EXTREMELY valuable for grants.

Malaysia’s Health Ministry itself acknowledged the influence of social media creators:

The Ministry of Health announced plans to engage food influencers to promote healthy eating and healthier lifestyles among Malaysians.

This is HUGE because it validates:

- health influencers ARE shaping behavior,
 - government recognizes this trend,
 - your narrative is aligned with national direction.
-

6. The BEST Narrative Insight

This is your strongest storyline:

Malaysians WANT healthier lifestyles,

BUT:

- nutrition guidance is fragmented,
- meal tracking is difficult,
- and clinical monitoring remains infrequent.

That becomes GlucoLens' opening.

7. Strong Problem Statement You Can Use

Here's a polished version:

Malaysia is experiencing a growing wave of digital health awareness driven by social media platforms such as TikTok and Instagram, particularly among younger and urban adults. TikTok alone recorded over 10 million downloads in Malaysia in 2024, while nutrition and wellness influencers now dominate a large portion of health-related content online.

However, despite increasing awareness, only 30.6% of Malaysian adults practice multiple healthy lifestyle behaviors consistently. Many consumers struggle to translate generalized online wellness advice into practical dietary decisions for Malaysian meals.

At the same time, dietitians face limited visibility into patients' daily eating habits between consultations, relying heavily on self-reported food recall and infrequent follow-ups. GlucoLens addresses this gap by transforming everyday meal photos into structured clinical nutrition insights using localized Malaysian food-recognition AI.

8. VERY IMPORTANT Strategic Advice

Do NOT invent fake statistics like:

✗ “70% of dietitians struggle...”

Judges/investors can easily challenge it.

Instead:

- use real workflow problems,
- cite NHMS,
- cite MOH,
- cite TikTok penetration,
- cite nutrition influencer trends.

That is MUCH stronger and more credible.

9. Strong One-Liner for Pitch Deck

“Social media has successfully increased health awareness among Malaysians — but consumers and clinicians still lack localized tools that transform everyday meals into actionable nutritional insights.”

That's a VERY strong commercialization narrative.